

CASE 27

LEUKOCYTE ADHESION DEFICIENCY

The traffic of white blood cells.

Newly differentiated blood cells continually enter the bloodstream from their sites of production: red blood cells, monocytes, granulocytes, and B lymphocytes from the bone marrow, and T lymphocytes from the thymus. Under ordinary circumstances, red blood cells spend their entire lifespan of 120 days in the bloodstream. However, white blood cells (leukocytes) are destined to emigrate from the blood to perform their effector functions. Lymphocytes recirculate through secondary lymphoid tissues, where they are detained if they encounter an antigen to which they can respond; macrophages migrate into the tissues as they mature from circulating monocytes; effector T lymphocytes and large numbers of granulocytes are recruited to extravascular sites in response to infection or injury. For example, it is estimated that each day 3 billion neutrophils enter the oral cavity, the most contaminated site in our body.

The process by which leukocytes migrate from the bloodstream to sites of infection is fairly well understood and involves interactions between adhesion molecules on the leukocyte surface and those on endothelial cells (Fig. 27.1). First, leukocyte flow is retarded by interactions between selectins, whose expression is induced on activated vascular endothelium, and fucosylated glycoproteins (for example sialyl-Lewis^x) on the leukocyte surface. Tight binding of leukocytes to the endothelial surface is then triggered by the actions of chemokines, such as CXCL1, CXCL2, and IL-8, which activate the leukocyte integrins (for example, LFA-1 and Mac-1) to adhere more tightly to their receptors. Crossing the endothelial cell wall also involves interactions between the leukocyte integrins and their receptors, while the subsequent direction of migration follows a concentration gradient of chemokines produced by cells already at the site of infection or injury. The process by which lymphocytes home to secondary lymphoid tissue is very similar, except that it is initiated by mucin-like addressins on lymphoid venules binding to L-selectin on the surface of naive lymphocytes (Fig. 27.2). The various adhesion molecules involved in leukocyte trafficking and other important leukocyte cell-cell interactions are described in Fig. 27.3.

An excellent opportunity to study the role of integrins is provided by a genetic defect in CD18, the common β chain of the three β_2 integrins: LFA-1 (CD11a:CD18),

TOPICS BEARING ON THIS CASE:

Complement receptors

Migration and homing of leukocytes

Phagocytic cell defects

Adhesion molecules in T-cell interactions

Bone marrow transplantation

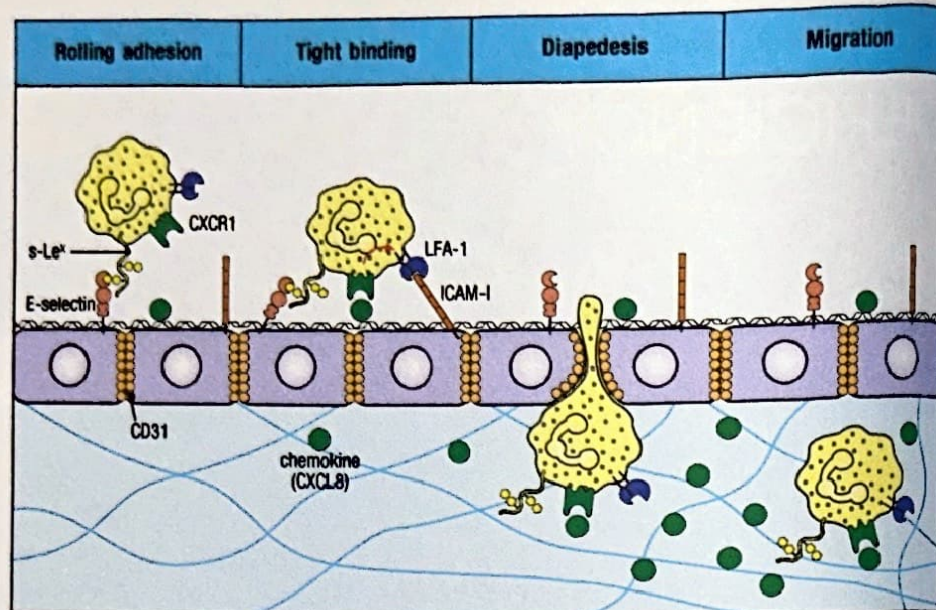
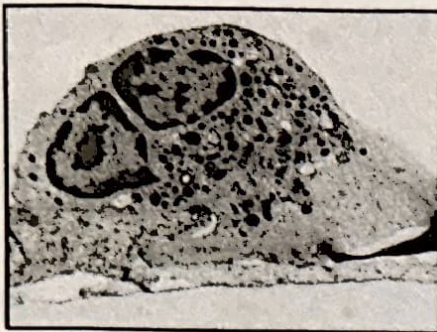
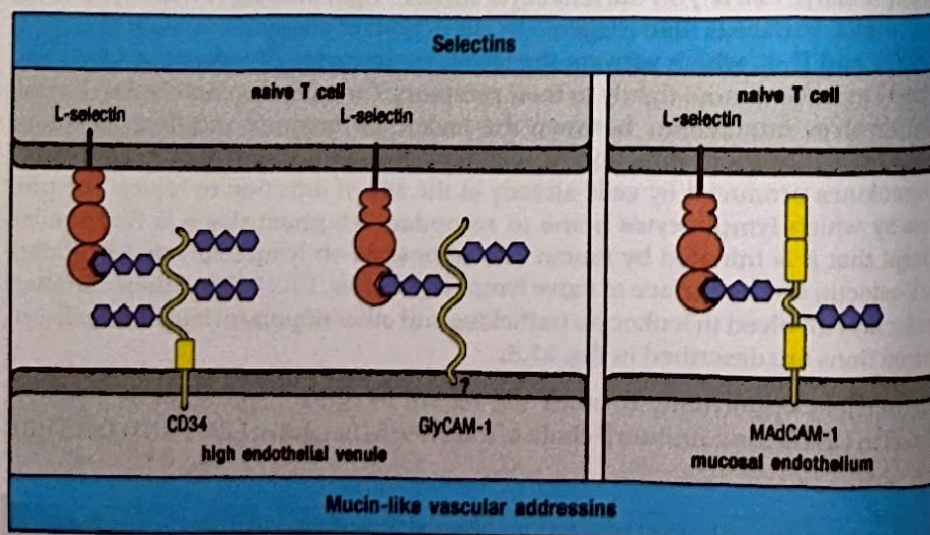


Fig. 27.1 Phagocytic leukocytes and effector T leukocytes are directed to sites of infection through interactions between adhesion molecules induced by chemokines. The first step (right panel) involves the reversible binding of leukocytes to vascular endothelium through interactions between selectins induced on the endothelium and their carbohydrate ligands on the leukocyte, shown here for E-selectin and its ligand the sialyl-Lewis^x moiety (s-Le^x). This interaction cannot anchor the cells against the shearing force of the flow of blood; instead, they roll along the endothelium, continually making and breaking contact. The binding does, however, allow stronger interactions, which occur as a result of the induction of the adhesion molecule ICAM-1 on the endothelium and the activation of its receptors, the integrins LFA-1 and Mac-1 (not shown) on the leukocyte. This binding is mediated by chemokines (such as CXCL8, which is recognized by the receptor CXCR1 on the leukocyte) retained on heparan sulfate proteoglycans on the endothelial cell surface.

Tight binding between these molecules arrests the rolling and allows the leukocyte to extravasate (leave the bloodstream) by squeezing between the endothelial cells forming the wall of the blood vessel (a process known as diapedesis). The leukocyte integrins LFA-1 and Mac-1 are required for extravasation and for migration toward chemoattractants. Adhesion between molecules of CD31, expressed on both the leukocyte and the junction of the endothelial cells, is also thought to contribute to diapedesis. Finally, the leukocyte migrates along a concentration gradient of chemokines (here shown as CXCL8) secreted by cells at the site of infection. The electron micrograph shows a neutrophil that has just started to migrate between two endothelial cells (bottom of photo). Note the pseudopod that the neutrophil has inserted between adjacent endothelial cells. The dark mass at the bottom right is an erythrocyte that has become trapped underneath the neutrophil. Photograph (×5500) courtesy of I. Bird and J. Spragg.

Fig. 27.2 L-selectin and the mucin-like vascular addressins direct the homing of naive lymphocytes to lymphoid tissues. L-selectin is expressed on naive T cells, which bind to the vascular addressins CD34 and GlyCAM-1 on high endothelial venules to enter lymph nodes. The relative importance of CD34 and GlyCAM-1 in this interaction is not clear. GlyCAM-1 is expressed exclusively on high endothelial venules but has no transmembrane region and it is unclear how it is attached to the membrane; CD34 has a transmembrane anchor and is expressed on endothelial cells but not exclusively on high endothelial venules. The addressin MAdCAM-1 is expressed on mucosal endothelium and guides entry into mucosal lymphoid tissue. L-selectin recognizes carbohydrate moieties on the vascular addressins.

Mac-1 (CD11b:CD18, which is the complement receptor CR3), and gp150,95 (CD11c:CD18, which is the complement receptor CR4). Children with this genetic defect suffer from leukocyte adhesion deficiency (LAD). They have recurrent pyogenic infections and problems with wound healing, and if they survive long enough they develop severe inflammation of the gums (gingivitis). Surprisingly,



		Name	Tissue distribution	Ligand
Selectins		P-selectin (PADGEM, CD62P)	Activated endothelium and platelets	PSGL-1, sialyl-Lewis ^x
		E-selectin (ELAM-1, CD62E)	Activated endothelium	Sialyl-Lewis ^x
Integrins		$\alpha_L\beta_2$ (LFA-1, CD11a:CD18)	Monocytes, T cells, macrophages, neutrophils, dendritic cells, NK cells	ICAM-1, ICAM-2
		$\alpha_M\beta_2$ (CR3, Mac-1, CD11b:CD18)	Neutrophils, monocytes, macrophages, NK cells	ICAM-1, iC3b, fibrinogen
		$\alpha_X\beta_2$ (CR4, p150.95, CD11c:CD18)	Dendritic cells, macrophages, neutrophils, NK cells	iC3b
		$\alpha_5\beta_1$ (VLA-5, CD49d:CD29)	Monocytes, macrophages	Fibronectin
Immunoglobulin superfamily		ICAM-1 (CD54)	Activated endothelium, activated leukocytes	LFA-1, Mac1
		ICAM-2 (CD102)	Resting endothelium, dendritic cells	LFA-1
		VCAM-1 (CD106)	Activated endothelium	VLA-4
		PECAM (CD31)	Activated leukocytes, endothelial cell-cell junctions	CD31

children with LAD are not unduly susceptible to opportunistic infections. This implies normal T-cell function despite the absence of LFA-1, which was thought to be important for T-cell adhesion to antigen-presenting cells (Fig. 27.4). The capacity to form antibodies is also unimpaired, showing that adequate collaboration between T cells and B cells can also occur without LFA-1. Finally, because neutrophils are released as normal from the bone marrow into the bloodstream but their migration into tissues is impaired, patients with LAD have a remarkable leukocytosis with neutrophilia.

Fig. 27.3 Adhesion molecules in leukocyte interactions. Several structural families of adhesion molecules play a part in leukocyte migration, homing, and cell-cell interactions: the selectins, mucin-like vascular addressins, the integrins, proteins of the immunoglobulin superfamily, and the protein CD44. The figure shows schematic representations of an example from each family, a list of other family members that participate in leukocyte interactions, their cellular distribution, and their partners in adhesive interactions. The nomenclature of the different molecules in these families is confusing because it often reflects the way in which the molecules were first identified rather than their related structural characteristics. Thus, although all the ICAMs are immunoglobulin-related, and all the VLA molecules are β_1 integrins, the CD nomenclature reflects the characterization of leukocyte cell-surface molecules by raising monoclonal antibodies against them, and embraces adhesion molecules in all the structural families. The LFA molecules were defined through experiments in which cytotoxic T-cell killing could be blocked by monoclonal antibodies against cell-surface molecules on the interacting cells, and there are LFA molecules in both the integrin and the immunoglobulin families. Alternative names for each of the molecules shown are given in parentheses. Sialyl-Lewis^x, which is recognized by P- and E-selectin, is an oligosaccharide present on cell-surface glycoproteins of circulating leukocytes. The ligand DC-SIGN, which binds to ICAM-3, is expressed on dendritic cells. It is suspected to have a major role in interactions between dendritic cells and T cells, and serves as a co-receptor for HIV.

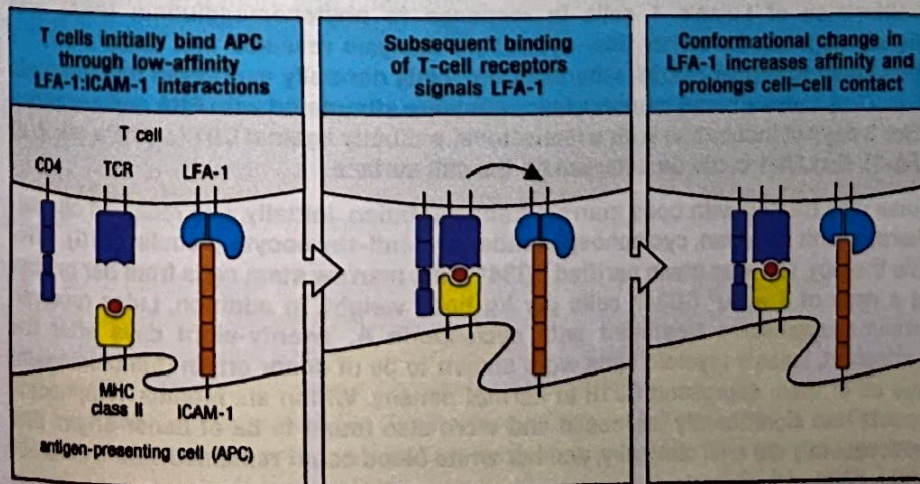


Fig. 27.4 Transient adhesive interactions between T cells and antigen-presenting cells are stabilized when specific antigen recognition induces high-affinity binding by LFA-1. When a T cell binds to its specific ligand on an antigen-presenting cell (APC), intracellular signaling through the T-cell receptor (TCR) induces a conformational change in LFA-1 that causes it to bind with higher affinity to ICAM-1 on the antigen-presenting cell. The cell shown here is a CD4 T cell.

The case of Luisa Ortega: a problem of immobile white blood cells.

Luisa, 4 weeks old, presents with acute omphalitis.

Family history of overwhelming infection.

Luisa Ortega was born at full term and weighed 3.7 kg. She was the second child born to the Ortegases. At 4 weeks of age, Luisa was taken by her parents to her pediatrician because she had swelling and redness around the umbilical cord stump (omphalitis), and a fever of 39°C. Her white blood cell count was $71,000 \mu\text{l}^{-1}$ (normal $5000\text{--}10,000 \mu\text{l}^{-1}$). She was treated in the hospital with intravenous antibiotics for 12 days and then discharged home with oral antibiotics. At the time of discharge her white blood count was $20,000 \mu\text{l}^{-1}$. Cultures obtained from the inflamed skin about the umbilical stump before antibiotic treatment grew *Escherichia coli* and *Staphylococcus aureus*.

The Ortegases had had a baby boy 3 years before Luisa's birth. At 2 weeks of age he developed a very severe infection of the large intestine (necrotizing enterocolitis). Separation of his umbilical cord was delayed. He subsequently suffered from multiple skin infections and he died of staphylococcal pneumonia at 1 year of age. Just before his death his white blood cell count was recorded at $75,000 \mu\text{l}^{-1}$.

Because of the previous family history, Luisa was referred to the Children's Hospital. At the time of her admission to the Children's Hospital she seemed normal on physical examination, and radiographs of the chest and abdomen were normal.

Cultures of urine, blood, and cerebrospinal fluid were negative. Her white blood count was $68,000 \mu\text{l}^{-1}$ (very high). Of her white cells, 73% were neutrophils, 22% lymphocytes, and 5% eosinophils (this distribution of cell types is in the normal range but the absolute count for each is abnormally high). Her serum IgG concentration was 613 mg dl^{-1} (normal), her IgM was 89 mg dl^{-1} (normal), and her IgA 7 mg dl^{-1} (normal). The concentration of complement component C3 in her serum was 185 mg dl^{-1} and that of C4 was 28 mg dl^{-1} (both normal).

A Rebuck skin window was performed. In this procedure, the skin of the forearm is gently abraded with a scalpel blade and a coverslip is placed on the abrasion. After 2 hours the coverslip is removed and replaced by another coverslip every subsequent 2 hours for a total of 8 hours. In this way, the migration of immune cells into the damaged skin can be monitored. In Luisa's case, no white cells accumulated on the coverslips. All types of leukocytes, however, were present in abnormally high numbers in her blood. Of her blood lymphocytes, 53% ($7930 \mu\text{l}^{-1}$) were CD3⁺ T cells (of these, 36% were CD4⁺ and 16% were CD8⁺, which are normal proportions), 25% ($3740 \mu\text{l}^{-1}$) were CD19⁺ B cells, and 14% were NK cells (as measured by CD16 expression). B-cell and NK-cell counts were both elevated.

Proliferation of Luisa's T cells in response to phytohemagglutinin (PHA) was slightly depressed. Further flow-cytometric analysis revealed that Luisa's CD3⁺ T cells failed to express CD18, whereas CD3⁺ cells normally expressed this molecule (Fig. 27.5). Luisa's blood mononuclear cells were stimulated with PHA and examined after 3 days of incubation with a monoclonal antibody against CD11a (the α chain of LFA-1). No LFA-1 could be detected on the cell surface.

Luisa was treated with bone marrow transplantation. Initially, she received chemotherapy with busulfan, cyclophosphamide, and anti-thymocyte globulin (ATG). After this therapy, she was given purified CD34⁺ bone marrow stem cells from her mother at a dose of 5×10^6 CD34⁺ cells per kg body weight. In addition, Luisa received immunosuppressive treatment with cyclosporin A. Twenty-eight days after the transplant, Luisa's myeloid cells were shown to be of donor origin (full chimerism) and all of them expressed CD18 at normal density. Within six months, lymphocyte counts had significantly increased and were also found to be of donor origin. She subsequently did well clinically, and her white blood count remained stable at about $5000\text{--}7800 \mu\text{l}^{-1}$.

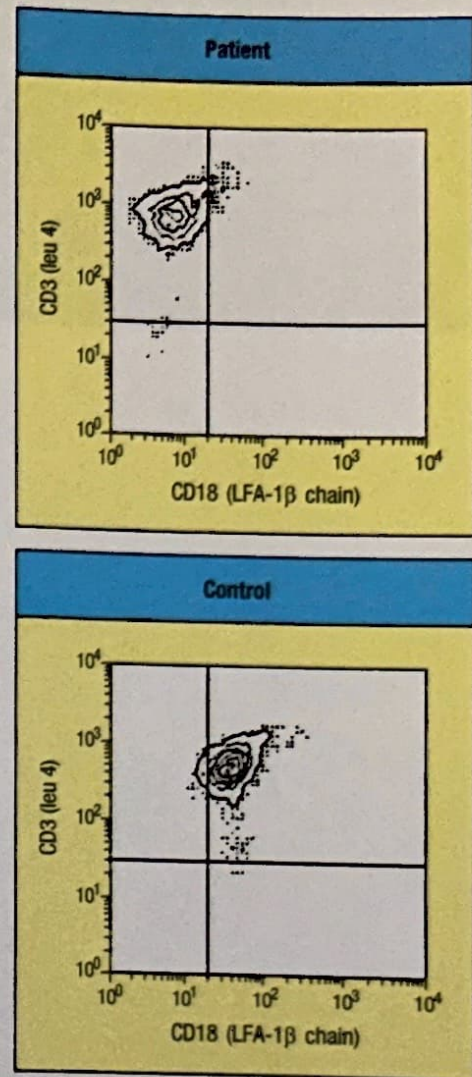
Fig. 27.5 Flow cytometric analysis of CD18 expression on Luisa's lymphocytes (upper panel) compared with that of a control subject (lower panel). The vertical axis measures fluorescence from labeled antibody against CD3; the horizontal axis measures fluorescence from labeled antibody against CD18. In the control subject, CD3⁺ cells also express CD18; in contrast, Luisa's CD3⁺ cells fail to express CD18 molecules.

Leukocyte adhesion deficiency (LAD).

Children such as Luisa, who lack the leukocyte integrin subunit CD18, are subject to recurrent, severe bacterial infections that are eventually fatal if untreated. Bone marrow transplantation has been very successful in rescuing infants with LAD from certain death. In such patients, encapsulated bacteria are coated as usual with antibody and complement. However, the neutrophils and monocytes that would normally be recruited to the site of infection are trapped in the bloodstream and cannot emigrate into the tissues because they lack the CD18-containing integrins LFA-1 and Mac-1/CR3. In a normal individual, the first coverslip in a Rebuck skin window contains many neutrophils. Monocytes begin to appear at 4 hours, and by 8 hours the coverslip contains predominantly monocytes and very few neutrophils. In Luisa's case the coverslips had no cells because her leukocytes were unable to emigrate from the bloodstream and onto the coverslip. For this reason, the levels of white cells in the bloodstream were very high—a characteristic of LAD. The ability to deal with pyogenic bacteria is further compromised because of the vital role of CR3-mediated uptake of these opsonized bacteria by neutrophils. The role of CR4, the third member of the β_2 -integrin family, is less well understood but, like CR3, it binds fragments of complement component C3, and it is thought to have a role in the uptake of bacteria by macrophages.

The importance of the daily, massive neutrophil emigration into the oral cavity is well illustrated by individuals with LAD, who, when they survive, invariably develop severe gingivitis. A poorly understood consequence of the lack of leukocyte emigration is the failure to heal wounds. Delayed separation of the umbilical cord is the earliest manifestation of this defect in wound healing. Subsequently, affected children may develop fistulas (abnormal connecting channels) in their intestine after bacterial infections of the gut.

In addition to CD18 deficiency (LAD type 1), other rarer forms of LAD have been identified. LAD type 2 is due to defects in fucose transport that impair selectin-mediated leukocyte rolling. LAD type 3 is due to mutations in the protein Kindlin-3, which regulates integrin-mediated signaling. Finally, deficiency of Rac2, a small GTPase, affects the expression of L-selectin, the generation of superoxide, actin polymerization, and chemotaxis, and shows combined features of LAD and chronic granulomatous disease (see Case 26).



Questions.

- 1 Can you deduce the inheritance pattern of LAD type 1 from this case?
- 2 In some families with LAD, a mild phenotype is observed and leukocytes of affected individuals express less than 10% of the normal amount of CD18. The mild phenotype is usually due to a splice site defect in the gene encoding CD18, whereas, in the severe form of the disease, nonsense or frameshift mutations, or deletions of the CD18 gene, are identified. Do these observations help you to predict the likely effects of treatment with monoclonal antibodies against CD18? Can you suggest situations in which such treatment might be therapeutically useful?

- 3 Can you suggest why the homing of T cells to lymphoid tissue is normal in patients with LAD and why these individuals do not suffer from susceptibility to opportunistic infections?
- 4 What manifestation of defective wound healing occurred in this case?
- 5 LAD type 2 is caused by mutations in the protein GDP-fucose transporter 1, which is important in the transport of fucose into the Golgi apparatus. Can you explain why such a defect should cause LAD?
- 6 Shortly after Luisa received a bone marrow transplant from her mother it was ascertained that she had complete chimerism of myeloid and lymphoid cells. How was this ascertained?
- 7 Why was Luisa treated with busulfan, anti-thymocyte serum, and cyclophosphamide before the bone marrow transplant?
- 8 Why was the response of Luisa's T cells to PHA depressed?
- 9 How does LAD differ from chronic granulomatous disease (Case 26)?